

PHY199-RAP

Lecture 2 Assignment

Due 18-September-2026 at 11:00AM. You don't need to submit anything.

This is not graded. A subset of these questions will appear on the quiz on 18-September-2026.

Name: _____

UBIT login: _____

1. What do coal, gas, nuclear, hydro, geothermal, and wind power plants all have in common?
 - a. They all burn a fuel to create heat that turns a turbine.
 - b. They all release carbon dioxide.
 - c. They all spin a magnet inside coils of wire (a turbine) to generate electricity.
 - d. They all rely on falling water.

2. Which electricity-generating technology is described as the lone exception that produces current directly, with no moving parts?
 - a. Geothermal
 - b. Hydroelectric
 - c. Wind
 - d. Solar photovoltaic

3. The three stages of getting electricity from a plant to your home are, in order:
 - a. Distribution → transmission → generation
 - b. Generation → transmission → distribution
 - c. Transmission → generation → distribution
 - d. Generation → distribution → transmission

4. How much energy is generated instantaneously versus how much is stored in batteries or other storage?
 - a. >99% stored, <1% instantaneous.
 - b. About the same stored and instantaneous (50/50).
 - c. <1% stored, >99% instantaneous.

5. Relative to the ground, if you are motionless at the top of a rollercoaster, which is the most accurate statement?
 - a. You have a large kinetic energy and a low potential energy.

- b. You have about equal kinetic and potential energies.
 - c. You have a large potential energy and a low kinetic energy.
6. Hydropower ultimately converts energy from which of the following mechanisms?
- a. Nuclear fission and magnetism.
 - b. Reflection of sunlight off the moon.
 - c. Gravity due to the earth and thermal energy from the sun.
 - d. Gravity due to the moon.
7. Which of the following does NOT use a motor or crank to do mechanical work?
- a. Water mill.
 - b. Turbine.
 - c. Bar magnet.
8. What are the units of power?
- a. Watts
 - b. Watt-hours
 - c. Joules
 - d. Kilograms
 - e. Newtons
9. If a wire A has a higher resistance than wire B, and both are connected to the same power supply providing the same voltage, what is the relationship between the currents through wires A and B?
- a. The current through wire A is higher than wire B.
 - b. The current through wire A is less than wire B.
 - c. The currents are equal in wires A and B.
10. Why do electrons have a magnetic spin?
- a. They're tiny spherical balls and spin around in 3 dimensions.
 - b. They're point particles and magnetic spin is a fundamental quantum mechanical property of electrons.
 - c. Electrons don't have a magnetic spin at all.
11. Which of the following situations will NOT result in a generated voltage (electromotive force) in a wire loop?
- a. A wire loop spinning end-over-end within a constant magnetic field.
 - b. A stationary wire loop in a spinning magnetic field.
 - c. A stationary wire loop in a stationary magnetic field.
12. Long-distance power lines are kept at which voltage to minimize resistive losses?
- a. High voltage, low current.
 - b. Low voltage, high current.